and M2M markets. This is mainly due to the fact that stakeholders [mobile operators, OEMs, eSIM providers, mobile virtual network operators (MVNO), etc] are now well knowledgeable about what an eSIM is and how it can be used efficiently. As a result of the 'yes-we-can' attitude, eSIM has now been widely deployed in M2M markets such as automotive, smart meters, and vending machines. It is necessary to deploy new roles for eSIM in terms of 5G security, particularly in subscriber identity privacy, cyber resistance, 5G roaming, or even network authentication.

Architecture of eSIM

Presently, Samsung is adopting the SM-DP+ architecture, which is responsible for the creation, generation, management, and protection of resolution profiles at the input/request of the operator. eSIM manufacturer is responsible for the initial cryptographic configuration and security architecture of eSIM and is also the provider of eSIM products. The operator provides access and communication services to subscribers through its mobile network infrastructure.

GSMA defines two types of profile in eSIM architecture:

Provision Profile. This is a communication profile. Initially, it is stored in the eSIM when it is shipped.

Operation Profile. This is a communication profile for connecting enterprise servers or the Internet.

Advantages of eSIM

- Smartphones will become smarter
- Since device manufacturers won't have to accommodate a SIM slot in the phones, they will have more flexibility in terms of design. With SIM card embedded in the device's internal hardware, the circuit will theoretically shrink and phones will become thinner
- It helps to build smartwatches
- It would be a game-changer for international travellers
- Simpler device setup
- Tethered smartphones can operate independently.
- Provides enhanced communication between devices
- The range of devices under mobile communication can be increased
- It also enables people to switch operators quickly and get operated
- Lowers the risk of device theft
- No need to manufacture and distribute SIM cards separately
- These have the potential to eliminate roaming charges abroad
- It is inexpensive as compared to SIM cards
- Provides more security
- Provides mobile number portability, which allows customers to charge up without changing their mobile numbers
- Provides better waterproofing and protection to the circuit

Disadvantages of eSIM

- Highly beneficial only if the devices are mass-produced
- In eSIMs, users need to store details such as contacts, messages, and other credentials on the cloud
- For users who keep changing their mobile phones, SIM cards are more beneficial than eSIMs
- There is a possibility of eSIM card data being hacked from cloud hosting
- Worldwide seamless operations are not easy
- eSIM service providers have to work harder to retain or acquire new customers

Challenges

- It needs tough adoption
- To shift to this new technology, every SIM service provider should agree that eSIMs are the future. Then only smartphone manufacturers will follow this path
- Switching to this new technology is quite time-consuming
- eSIM brings the digitalisation of the SIM card production process and requires a greater level of activities by operators and their partners
- Operators need to have deeper knowledge about their eSIM profile to ensure these can support different devices as these are deployed around the world
- Many eSIM enabled consumer devices would demand new sign up onboarding experiences for mobile operators
- Without entanglement, an eSIM-enabled device will not be capable of taking advantage of new features such as multi-SIM support, VoLTE/Vo WiFi, or dynamic usage and subscription notifications

eSIM and service providers

Researchers have demonstrated how the use of eSIM can dramatically accelerate the growth and ecosystem of mobile phones. Unfortunately, today there are only a few mobile phone devices that are eSIM ready. The eSIM logo is required to help manufacturing companies to promote their eSIM-enabled products.